Here is the complete project documentation for your hospital billing and classification program, structured in the same format as before!

1. Problem Statement, Objectives, and Assumptions

Problem Statement

Manual calculation of medical bills involving multiple service fees, conditional demographic discounts (such as senior citizen concessions), and subsequent bill categorization can lead to billing errors and administrative overhead. A programmatic solution is needed to seamlessly compute, discount, and classify patient medical invoices.

Objectives

- **Automate Fee Aggregation:** To combine consultation, diagnostic, and treatment costs into a gross total.
- **Conditional Discounting:** To automatically apply a 10% discount for eligible senior citizen patients while keeping standard rates for normal patients.
- **Bill Classification:** To categorize the final payable amount into manageable financial tiers (Low Bill vs. High Bill) for hospital management.

Assumptions

- Financial inputs (consultation, diagnostic, treatment fees) are numeric values.
- Patient type is provided as text and handled case-insensitively.
- The senior citizen discount is calculated as a flat 10% of the gross total.

2. Algorithm, Flowchart, and Pseudocode

Algorithm

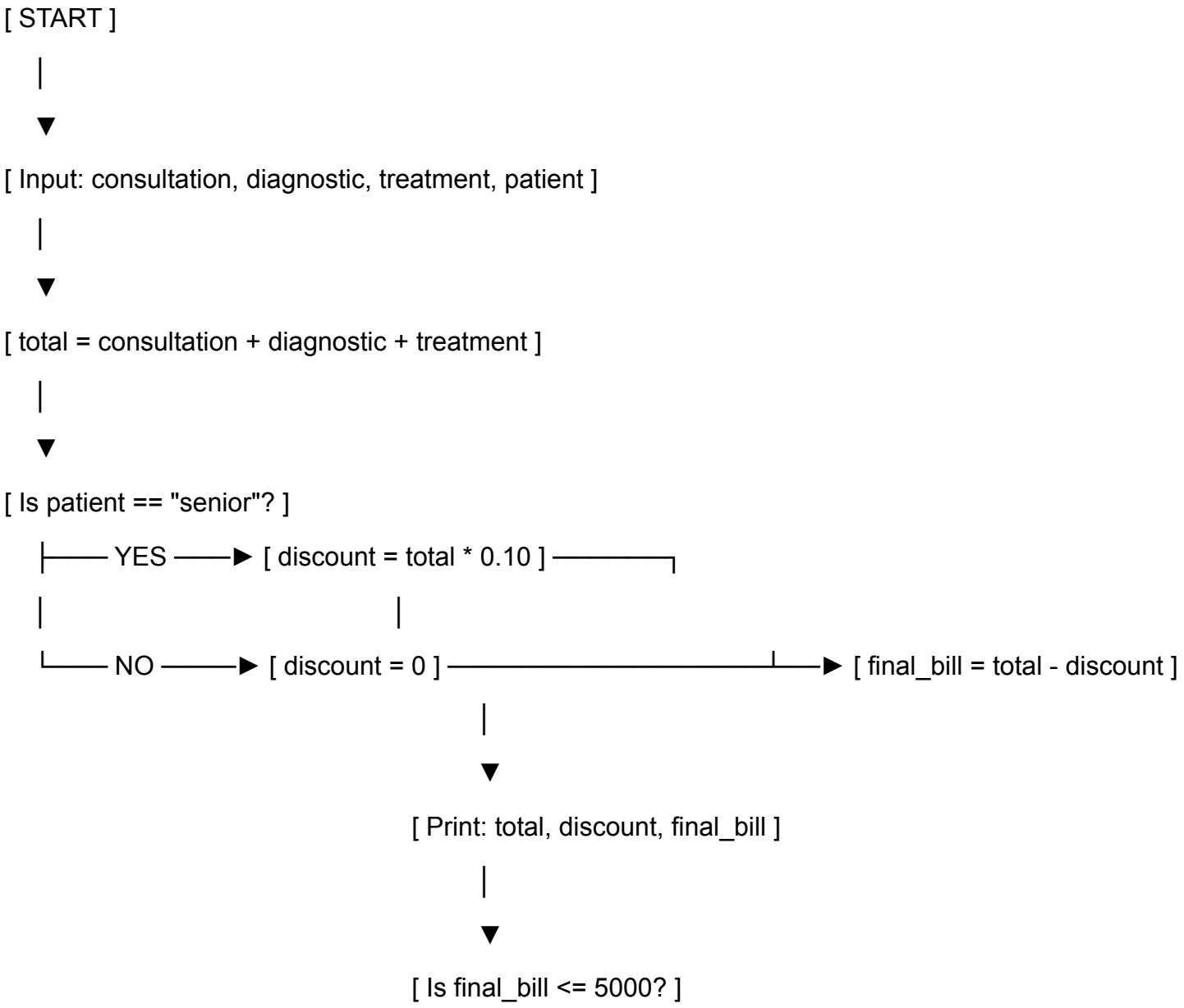
1. **Start** the program.
2. **Input** consultation fee, diagnostic fee, and treatment fee as floating-point numbers.
3. **Input** patient type as a string (senior or normal).
4. **Calculate Gross Total:** total = consultation + diagnostic + treatment.
5. **Evaluate Discount Condition:**
    - Convert patient type to lowercase.

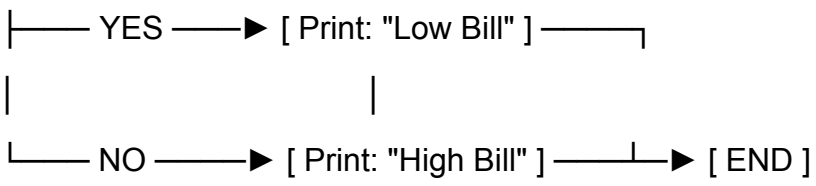
- If patient == "senior", compute discount = total * 0.10.
- Else, set discount = 0.

6. **Calculate Final Bill:** final_bill = total - discount.
7. **Display Results:** Print total, discount, and final bill amounts.
8. **Evaluate Classification Condition:**
   - If final_bill <= 5000, print "Classification: Low Bill".
   - Else, print "Classification: High Bill".
9. **End** the program.

Flowchart Representation

Plaintext


[START]
[Input: consultation, diagnostic, treatment, patient]
[total = consultation + diagnostic + treatment]
[Is patient == "senior"?]
YES
[discount = total * 0.10]
NO
[discount = 0]
[final_bill = total - discount]
[Print: total, discount, final_bill]
[Is final_bill <= 5000?]



Pseudocode

Plaintext

```
BEGIN
    PRINT "Enter consultation fee: "
    READ consultation
    PRINT "Enter diagnostic fee: "
    READ diagnostic
    PRINT "Enter treatment fee: "
    READ treatment
    PRINT "Enter patient type (senior/normal): "
    READ patient

    total = consultation + diagnostic + treatment

    IF lowercase(patient) == "senior" THEN
        discount = total * 0.10
```

```
    ELSE
        discount = 0
    ENDIF

    final_bill = total - discount

    PRINT "Total Bill: Rs.", total
    PRINT "Discount: Rs.", discount
    PRINT "Final Bill: Rs.", final_bill

    IF final_bill <= 5000 THEN
        PRINT "Classification: Low Bill"
    ELSE
        PRINT "Classification: High Bill"
    ENDIF
END
```

3. Python Source Code, Test Cases, and Outputs

Python Source Code

Python

```
# Hospital Billing and Classification System

consultation = float(input("Enter consultation fee: "))
diagnostic = float(input("Enter diagnostic fee: "))
treatment = float(input("Enter treatment fee: "))
patient = input("Enter patient type (senior/normal): ")

total = consultation + diagnostic + treatment

if patient.lower() == "senior":
    discount = total * 0.10
else:
    discount = 0

final_bill = total - discount

print("Total Bill: Rs.", total)
print("Discount: Rs.", discount)
print("Final Bill: Rs.", final_bill)
```

```
if final_bill <= 5000:
    print("Classification: Low Bill")
else:
    print("Classification: High Bill")
```

Test Cases Table

Test Case ID	Consultation	Diagnostic	Treatment	Patient Type	Total (Rs.)	Discount (Rs.)	Final Bill (Rs.)	Classification	Status
TC_01	500	1000	2000	normal	3500.0	0.0	3500.0	Low Bill	Pass
TC_02	1000	2500	3000	senior	6500.0	650.0	5850.0	High Bill	Pass
TC_03	800	1200	1500	Senior	3500.0	350.0	3150.0	Low Bill	Pass
TC_04	2000	3000	6000	normal	11000.0	0.0	11000.0	High Bill	Pass

Simulated Terminal Output

Bash

Enter consultation fee: 1000

Enter diagnostic fee: 2500

Enter treatment fee: 3000

Enter patient type (senior/normal): senior

Total Bill: Rs. 6500.0

Discount: Rs. 650.0

Final Bill: Rs. 5850.0

Classification: High Bill

4. Conclusion

This Python script effectively integrates numeric arithmetic, string manipulation (.lower()), and multi-tier conditional branching to solve a practical medical billing challenge. By automating discount calculations for senior citizens and organizing expenditures into clear financial brackets, the script serves as a robust prototype for healthcare administration software.